AI-Assisted Assessment Design in Mathematics and Statistics

A Bloom's taxonomy look at what my final exams asked of students

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Revise and resubmit, *Teaching Statistics*

The question

Most research on AI in education looks at students. I asked about the instructor's side.

Writing an exam item that truly needs reasoning takes time. One item with distractors that catch real misconceptions can take two or more hours. Under a heavy adjunct load, the easy default is a procedural item.

So my question:

1. When AI helps me write a final exam, does it change *what* the exam asks students to do, not only how fast I can write it?
2. If it does, in which direction?

A common worry is that AI lowers rigor. I looked at the items themselves.

Four final exams, one instructor

Course	Term	Exam	Enrolled
Introductory statistics	Winter 2025	Written without AI	29
	Winter 2026	Written with AI	28
College algebra (STEM)	Spring 2025	Written without AI	34
	Spring 2026	Written with AI	26

- A large urban open-access community college in California; both courses had a corequisite support section.
- The 2025 terms used no AI. In the AI terms I used Claude (Anthropic) to draft items, the formula sheet and the answer key.
- I reviewed, revised and approved every item. This is an observational comparison, not an experiment.

The method in plain words: paired Bloom's coding

I put each exam beside its partner from the year before and coded every item by the main thinking it asks for.

- **Remember:** recall a definition or a formula.
- **Understand:** read a graph or explain a concept.
- **Apply:** carry out a procedure with the values given.
- **Analyse:** judge competing conclusions, choose a method, or reason from output to a claim.

Statistics: the multiple-choice items only (34 before, 25 with AI). Algebra: all 20 items on each exam. When an item sat between two levels, I coded it at the higher one.

The method is designed so other instructors can repeat it on their own exams.

Same learning outcome, different demand

Winter 2025, without AI: Apply

A poll gives a sample and a percentage. Find the test statistic and the *p*-value.

Every value is given. Students run the formula.

Winter 2026, with AI: Analyse

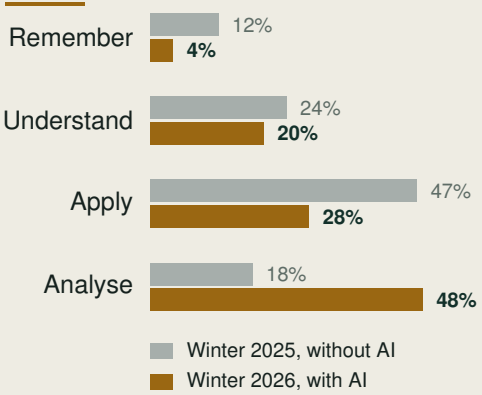
The *p*-value is given. Choose the correct decision and conclusion.

Three distractors carry real misconceptions, such as “accept H_0” or “reject H_a”.

A second pair from the paper. Before: compute a regression line from summary statistics. With AI: pick the one correct reading of a confidence interval out of four.

In the first pair the topic stayed. The thinking asked of students changed.

Statistics: the exam moved toward Analyse



Share of multiple-choice items at each level.

18%
Analyse, before

48%
Analyse, with AI

6 of 34 items before, 12 of 25 with AI: a gain of 30 percentage points.

Apply fell from 47% to 28%.

Analyse replaced Apply as the most common level on the statistics final.

Algebra: same demand, new structure

Level	Spring 2025	Spring 2026
Understand	5%	0%
Apply	85%	85%
Analyse	10%	15%

Analyse moved by a single item.

What changed was the structure:

- three labelled work steps under each problem;
- distractors built on known errors, across 19 multiple-choice items;
- a 2-page formula sheet and two parallel forms;
- a study guide and a practice final released in advance.

Bloom's levels alone would say "no effect". The structure changed almost everywhere.

Final exam means held or rose; the statistics pass rate dipped

Course	Term	Mean final exam score	Course pass rate
Statistics	Winter 2025, without AI	74.0%	93.1% (27/29)
	Winter 2026, with AI	75.7%	89.3% (25/28)
Algebra	Spring 2025, without AI	61.8%	67.6% (23/34)
	Spring 2026, with AI	78.7%	69.2% (18/26)

- Statistics: the mean held on a final with far more Analyse items; the pass rate fell from 93.1% to 89.3%.
- Means count students who sat the final: statistics 28 of 29 before; algebra 32 of 34 before, 19 of 26 with AI.
- I report these descriptively only. The exams are different instruments, and cohorts differ.

Was the new exam simply easier?

Three things make that unlikely as the main explanation:

1. The shift was toward *higher* demand. Analyse items doubled.
2. A formula sheet removes formula recall, not reasoning. A student who cannot choose the right test gains nothing from the sheet.
3. Performance held rather than jumped, which suggests the exam still separated students.

Without an equivalent-forms analysis, I cannot rule out that the AI-assisted exam was somewhat easier.

Easier may be part of it. The evidence does not support it as the main reason for the shift.

AI amplified my intentions; it did not replace my judgment

8 to 12
hours without AI

2 to 3
hours with AI

My estimate for writing 12 Analyse items with distractors that carry real misconceptions. My work moved from writing items to judging them.

- In statistics, my goal was reasoning over computation. AI made those items practical to write.
- In algebra, my goal was clear, diagnosable, fairly prepared procedural mastery. AI helped build that instead.

AI did what I asked. The choice of what to ask stayed with me.

What this study cannot show

- **One instructor, observational.** Results need replication across instructors and colleges.
- **Different exams.** The pairs are not equivalent instruments, and cohorts and terms differ.
- **One coder.** I coded every item myself; there was no inter-rater check.
- **No item-level scores** by Bloom's level.
- **One AI tool and one prompting approach.** Others may give different results.
- **Exam demand is not student learning.** I cannot say students reasoned more deeply.

What this means for practice

- **Name the Bloom's level in the prompt.** "Write five hypothesis testing questions" gives mostly Apply items. Asking for Analyse items with specific reasoning tasks gives different ones.
- **Treat the formula sheet as a design decision.** With AI, mine took about 45 minutes.
- **Use a separate real scenario for each part.** My statistics final used seven.
- **Code your own exams in pairs,** and look at structure too, not only levels.

The instructor's expertise is knowing which level to target and why.

The takeaway

With AI help, my statistics final asked for much more reasoning: Analyse items went from 18% to 48%, and the mean score held. My algebra final kept the same demand and gained structure.

In both courses, AI amplified what I was already trying to do. It removed the time barrier; it did not make the choices.

Thank you. Safaa Dabagh
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